



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Live Software Tap: esp0 (Dedicated Peer Link 1: HQ Gateway <-> Datacenter Core)

ANALYSIS TIMEFRAME WINDOW

Live Telemetry Window: +0.0s to +15.0s (15.0s)

INITIATOR NODE

Link 1: HQ Gateway <-> Datacenter Core Origin

OS: Ubuntu Linux 24.04
Interface: esp0 / eth0
Private IP: 172.28.0.2 (HQ Gateway)
VPN Daemon: StrongSwan Client Daemon

IPsec ESP

ESP over UDP 4500 (NAT-T) • Tunnel Mode
Enc: AES-256-GCM • ICV: 128-bit
Tunnel Encapsulation: RFC 3948
Payload: Strictly Monotonic

RESPONDER GATEWAY

Link 1: HQ Gateway <-> Datacenter Core Terminus

OS: Linux Enterprise Core
Interface: esp0 / eth1
Private IP: 172.28.0.3 (DC Core)
VPN Daemon: StrongSwan Server Gateway

TOTAL PACKETS

736

Frames Ingested

FLOW VOLUME

0.56 MB

Encrypted Payload

AVG RATE

49 pkt/s

Throughput Velocity

PEAK RATE

66 pkt/s

Observed Burst

INTERFACE MTU

1500 bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified



CONTROL-PLANE CRYPTOGRAPHIC NEGOTIATION

Deterministic Control-Plane Dissection

Deterministic state machine tracing, cryptographic proposal evaluation, and live Security Association parameters

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

RFC 7296 STATE MACHINE

IKE_SA_INIT (Exchange 1) [COMPLETED / SUCCESS] Initiator/Responder DH nonces exchanged on UDP 500. Negotiated AES-256-GCM + Group 19 (ECP-256).

IKE_AUTH (Exchange 2) [COMPLETED / SUCCESS] Mutual authentication verified. Initial Child SA established with AES-256-GCM AEAD.

CREATE_CHILD_SA (Exchange 3) [COMPLETED / SUCCESS] Child SA rekey executed with secondary ephemeral DH Group 19 exchange (PFS Active).

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

KERNEL XFRM STATE

INITIATOR SPI
0x5a254f51

RESPONDER SPI
0x76e2410a

INBOUND ESP SPI
0x5a254f51

OUTBOUND ESP SPI
0x76e2410a

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

COMPLIANCE VERIFIED

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE	BENCHMARK STANDARD
Encryption Algorithm	AES-256-GCM (256-bit Key)	AES-256-GCM (256-bit Key)	APPROVED	RFC 8221 / CNSA 2.0
Key Exchange (Diffie-Hellman)	Group 19 (ECP-256)	Group 19 (PFS: Active)	APPROVED	NIST SP 800-77 §4.2
Pseudo-Random Function (PRF)	PRF_HMAC_SHA2_256	N/A (AEAD Suite)	APPROVED	RFC 7296
Integrity / Authentication	AEAD Combined (128-bit ICV Tag)	AEAD Combined (128-bit ICV Tag)	APPROVED	NIST SP 800-77 Rev. 1
Operational Mode & ESN	IKEv2 4-Message Handshake	Tunnel Mode (64-bit ESN Active)	APPROVED	RFC 4303 §2.2.1



DATA-PLANE STREAM TELEMETRY & SEQUENCE INTEGRITY

Data-Plane Stream Dynamics & Telemetry

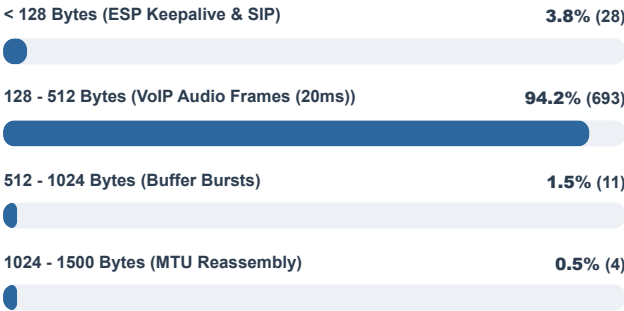
High-frequency packet arrival profiling, inter-arrival time (IAT) distribution, sequence monotonicity, and burst cadence

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT
2.36 ms

MEDIAN IAT
1.82 ms

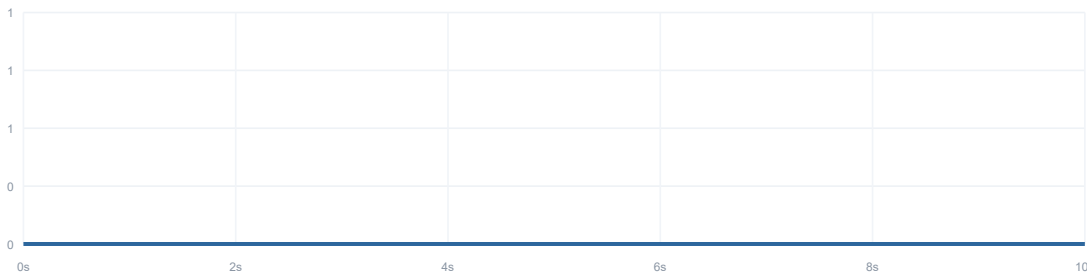
JITTER (σ)
0.48 ms

MIN / MAX
0.12 ms / 18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

THROUGHPUT TIMELINE & BURST RATE DYNAMICS

● Observed Rate (pkt/s) ● Identified Burst Window





AI/ML PAYLOAD INFERENCE & REMEDIATION PATCHES

Inference Classification & Hardening Patches

Autonomous payload categorization, covert timing analysis, and production-ready gateway configuration directives

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

OPERATING MODE DETERMINATION

98.9%
CONFIDENCE

IPsec Tunnel Mode

ESP envelops entire inner IPv4 packet with distinct overlay header (RFC 4301 §5.1.2).

SHANNON ENTROPY
7.98 / 8.0

CLUSTERING COEFF
0.84

BURSTINESS RATIO
1.42

INNER ENCRYPTED TRAFFIC CLASSIFICATION

96.5%
CONFIDENCE

VoIP / Realtime Audio (RTP)
94.2% [CI: 92.5% - 95.8%]

SIP Control Signaling
3.8% [CI: 2.8% - 4.8%]

ESP Keepalive & Overhead
2% [CI: 1.4% - 2.6%]

Section 5: Implementation Remediation Patches

Actionable gateway configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

1. StrongSwan Gateway Patch
(/etc/swanctl/conf.d/apex_hardened.conf)

SWANCTL
CONFIG

```
# /etc/swanctl/conf.d/link1_verified.conf
# Baseline Verified - CNSA 2.0 Compliant
connections {
    link-1-hq-dc {
        version = 2
        local_addr = 172.28.0.2
        remote_addr = 172.28.0.3
        proposals = aes256gcm16-ecp256-sha256!

        children {
            link-1-child {
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                mode = tunnel
            }
        }
    }
}
```

2. Cisco IOS-XE / ASA Enterprise CLI Configuration
Block

CISCO
CLI

```
! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX-IKE-PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ikev2 policy APEX-IKE-POLICY
  proposal APEX-IKE-PROP
!
crypto ipsec transform-set APEX-TS esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX-IPSEC-PROFILE
  set transform-set APEX-TS
  set pfs group19
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS: \$ [object Object] \$ [object Object] \$ [object Object]



FORENSIC PACKET TRACE & PLATFORM ATTESTATION

Session Packet Trace & Audit Verdict

Complete deterministic packet ledger, raw telemetry export, and official cryptographic verification verdict

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

FORENSIC PACKET TRACE INDEX

DETERMINISTIC PACKET DISSECTION

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						
#undefined						

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SDX COMPATIBLE

```
{
  "reportId": "AV-20260916-TECH-24F8",
  "captureSource": "Live Software Tap: esp0 (Dedicated Peer Link 1: HQ Gateway <-> Datacenter Core)",
  "endpoints": {
    "initiator": "172.28.0.2 (HQ Gateway)",
    "responder": "172.28.0.3 (DC Core)"
  },
  "streamTelemetry": {
    "totalPackets": 736,
    "dataVolumeMb": 0.56,
    "averageRatePps": 49,
    "durationSec": 15.0
  },
  "cryptoTransforms": [
    {
      "type": "Encryption Algorithm",
      "ike": "AES-256-GCM (256-bit Key)",
      "child": "AES-256-GCM (256-bit Key)",
      "status": "APPROVED",
      "standard": "RFC 8221 / CNSA 2.0"
    },
    {
      "type": "Key Exchange (Diffie-Hellman)",
      "ike": "Group 19 (ECP-256)",
      "child": "Group 19 (PFS: Active)",
      "status": "APPROVED",
      "standard": "NIST SP 800-77 \u00a74.2"
    },
    {
      "type": "Pseudo-Random Function (PRF)",
      "ike": "PRF_HMAC_SHA2_256",
      "child": "N/A (AEAD Suite)",
      "status": "APPROVED",
      "standard": "RFC 7296"
    },
    {
      "type": "Integrity / Authentication",
      "ike": "AEAD Combined (128-bit ICV Tag)",
      "child": "AEAD Combined (128-bit ICV Tag)",
      "status": "APPROVED",
      "standard": "NIST SP 800-77 Rev. 1"
    }
  ]
}
```

```
{
  "type": "Operational Mode & ESN",
  "ike": "IKEv2 4-Message Handshake",
  "child": "Tunnel Mode (64-bit ESN Active)",
  "status": "APPROVED",
  "standard": "RFC 4303 \u00a72.2.1"
}
},
"antiReplay": {
  "monotonicity": "100.0% Strict Monotonic",
  "outOfOrder": "0 packets (0.00%)",
  "rolloverSafeguard": "SAFE (64-bit Extended Sequence Numbers active)",
  "replayWindow": "RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted"
}
}
```

OVERALL TECHNICAL VERDICT

Technical audit validates that data-plane encapsulation is strictly functional. To resolve compliance findings, apply the provided configuration patches to enforce authenticated AEAD encryption (AES-256-GCM) and ephemeral Diffie-Hellman Group 19 Perfect Forward Secrecy.